Test of Mathematics for University Admission

DELUXE

Set 1 · Paper 1

20 questions · 75 minutes · no calculator

A paper by

Viluxent (mkb6)

Twenty questions written as a prediction of the October 2026 Paper 1, built from the 2024, 2025 and 2026 papers rather than to a target difficulty. The topic spread, the option counts, the length of the stems and the balance between routine and awkward are taken from those papers. Full worked solutions and an answer key follow the paper.

Before you begin

This is a prediction paper. It is not a collection of hard problems, and it is not trying to be harder than the real thing. Every question was written after reading the 2024, 2025 and 2026 papers and asking one narrow question: what is this examination actually going to put in front of you in October?

So the topic spread is theirs, not ours. Indices and logarithms, algebra and polynomials, sequences and series, coordinate and circle geometry, trigonometry, radian measure and mensuration, differentiation, integration, functions and graphs, probability, statistics, counting and number all appear, in roughly the proportions those papers use. The stems are as terse as theirs. Two or three questions here are harder than anything you will meet on the day; two or three are easier. That is deliberate. A real paper is not a ramp, and an easy question can sit at 18.

What the paper does not do is reward grinding. Each question has a short route once the key idea is found, and the options have been written so that substituting numbers into them tells you nothing. Recognising an option is not evidence that you have solved anything: in most questions the distractors are exactly the values produced by the standard mistakes — solving for the wrong variable, taking only one branch of a case split, forgetting that an endpoint is excluded, stopping a telescoping sum one term early, or answering the question that was asked one step earlier.

Sit it properly: seventy-five minutes, no calculator, no notes, in one go. Then mark it against the table overleaf, which converts your raw mark to the TMUA's own scale. The number you get is meant to be an honest estimate of what you would score on the day. Treat it as an estimate, not as a verdict.

Then, and this is the part almost everybody skips, go back to every question you got wrong and rewrite the solution yourself from a blank page, without looking. If you cannot reproduce it, you have not learned it yet.

A word on everything else. Do the past papers, all of them, twice. Work through every free resource you can find from the people who genuinely make good TMUA material. There is far more of it now than there has ever been, and it costs you nothing but your time.

And if something in here breaks you for an afternoon, that is the paper working. The questions that cost you the most are the ones you will still be able to do in October.

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Candidate instructions

- You have 75 minutes to attempt all 20 questions.
- No calculator may be used. No formula sheet is provided or needed.
- Each question has exactly one correct answer among the options given. The number of options varies between questions, so read each list carefully before answering.
- There is no penalty for a wrong answer, so attempt everything.
- Only the option you select is marked; use the surrounding space for rough working.

A note on style

This is a Paper 1. Every question has a single numerical or algebraic answer, and none of them is a reasoning item: that is what Paper 2 is for, and the recent Paper 1s carry essentially no necessary-and-sufficient work. Most questions here combine two areas of the specification, which is the one liberty this set takes with the real thing, and it is taken because the harder questions on the recent papers do the same.

The options have been chosen adversarially. In most questions the distractors are the values produced by the standard mistakes, so recognising an option tells you nothing. Where an answer is a surd or a fraction, the neighbouring options are the surds and fractions that the wrong route actually produces, not decorative alternatives.

Score calibration

Mark one point per question and read your raw mark across. The TMUA reports a score on a scale from 1.0 to 9.0, and in the middle of the range the conversion is close to linear.

17–20	scaled 8.0 to 9.0 — the top few per cent of the cohort
14–16	scaled 7.0 to 7.9 — clear of every published offer threshold
11–13	scaled 6.0 to 6.9 — the 6.5 borderline runs through this band
8–10	scaled 5.0 to 5.9 — the middle of the national cohort
0–7	below 5.0 — work through the solutions before you sit the next one

The conversion is indicative: the TMUA publishes its own raw-to-scaled table only after each sitting, and it moves by a few tenths from year to year with the difficulty of the paper and the strength of the cohort.

The real numbers x and y are both greater than 0, with $x > 1$, and

$$x^{\log_3 y} = 27, \qquad (xy)^{\log_3 x} = 3^{13}.$$

Find $\log_3(xy)$.

A $\dfrac{13}{10}$

B 3

C $\sqrt{13}$

D $\sqrt{10}$

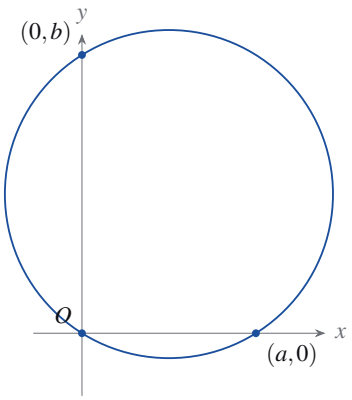
E $\dfrac{3\sqrt{10}}{10}$

F $\dfrac{13\sqrt{10}}{10}$

Question 2

A circle passes through the origin O, meets the positive x-axis again at $(a,0)$ and meets the positive y-axis again at $(0,b)$, where a and b are integers with $0 < a < b$. The radius of the circle is $\frac{65}{2}$.

How many such circles are there?



A 4

B 5

C 6

D 8

E 12

F 16

Question 3

The polynomial

$$P(x) = x^3 + ax^2 + bx + c$$

has three roots, all of which are positive integers, not necessarily distinct. The coefficients satisfy

$$a + b + c = -2026.$$

Find the largest possible value of a.

A -2030

B -50

C -46

D -42

E -39

F 42

Question 4

The sequence $u_1, u_2, u_3, \ldots$ of positive numbers satisfies $u_1 = 1$ and

$$u_1 + u_2 + \cdots + u_n = \frac{n(n+2)}{3}\, u_n \qquad \text{for every } n \geq 1.$$

Find $\sum_{n=1}^{\infty} u_n$.

A $\frac{12}{5}$

B 4

C 3

D $\frac{18}{5}$

E $\frac{24}{5}$

F 6

Find the number of values of x with $0 \leq x \leq 2\pi$ for which

$$\sin^6 x + \cos^6 x = \tfrac{1}{4}\left(\sin 2x + 1\right).$$

A 0

B 1

C 2

D 3

E 4

F 6

G 8

Question 6

Evaluate

$$\int_{-2}^{2} \frac{x^4+1}{1+3^x}\,dx.$$

A $\frac{21}{5}$

B $\frac{32}{5}$

C $\frac{42}{5}$

D $\frac{52}{5}$

E $\frac{64}{5}$

F $\frac{84}{5}$

Question 7

A fair six-sided die, with faces numbered 1 to 6, is rolled repeatedly. The rolling stops as soon as either a 6 has been rolled, or two odd numbers have been rolled on consecutive rolls.

Find the probability that the rolling stops because a 6 has been rolled.

A $\frac{5}{12}$

B $\frac{1}{3}$

C $\frac{3}{7}$

D $\frac{4}{9}$

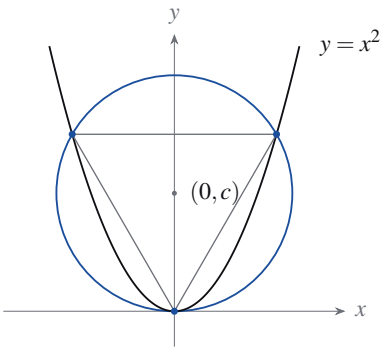
E $\frac{2}{5}$

F $\frac{5}{9}$

G $\frac{1}{2}$

The circle C has centre $(0, c)$, where $c > 0$, and passes through the origin. It meets the curve $y = x^2$ at exactly three points, and those three points are the vertices of an equilateral triangle.

Find c.



A $\frac{1}{2}$

B 1

C $\frac{3}{2}$

D 2

E $\frac{5}{2}$

F 3

The function f is defined for all real x by

$$f(x) = x|x| - 4x.$$

For those real numbers k for which the equation $f(x) = k$ has exactly three distinct real roots, let S be the sum of those three roots.

Find the set of all possible values of S.

A $-2 < S < 2$

B $-\sqrt{2} < S < \sqrt{2}$

C $2 - 2\sqrt{2} < S < 2\sqrt{2} - 2$

D $0 \leq S < 2\sqrt{2} - 2$

E $-2\sqrt{2} < S < 2\sqrt{2}$

F $-4 < S < 4$

Question 10

How many positive integers n are such that

$$n^2+n+1 \quad \text{divides} \quad n^3+272\,?$$

A 5

B 6

C 7

D 8

E 12

F 16

G infinitely many

Let $f(x) = x^3 - 3x^2 + k$, where k is a real constant. Find the set of values of k for which the equation

$$\left|f(x)\right| = 2$$

has exactly four distinct real solutions.

A $-6 < k < 2$

B $-2 < k < 2$

C $2 < k < 6$

D $-2 \leq k \leq 6$

E $-2 < k < 6$ and $k \neq 2$

F $-2 < k < 6$

Question 12

How many subsets S of $\{1,2,3,\ldots,15\}$ have the property that no element of S is equal to three times another element of S?

(The empty set counts as one such subset.)

A 2560

B 4096

C 6912

D 8640

E 10368

F 12960

G 17280

Question 13

Find the smallest positive integer N for which

$$\sum_{n=1}^{N} \frac{1}{n\sqrt{n+1}+(n+1)\sqrt{n}} > \frac{9}{10}.$$

A 10

B 49

C 81

D 99

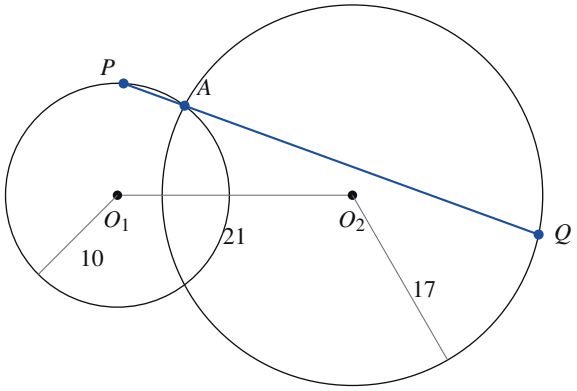
E 100

F 101

Question 14

Two circles have radii 10 and 17, and the distance between their centres is 21. They meet at two points, one of which is A. A straight line through A meets the first circle again at P and the second circle again at Q, with A lying between P and Q.

Find the greatest possible value of PQ.



A 16

B 21

C 27

D 34

E 42

F $2\sqrt{445}$

Question 15

A list of seven positive integers has mean 7 and median 7. The number 7 is the mode of the list, and it is the only mode.

Find the largest possible value of the range of the list.

A 18

B 23

C 20

D 22

E 24

F 41

Find the smallest possible value of

$$\frac{a^2+b^2+c^2}{ab+bc}$$

as a, b and c range over all positive real numbers.

A $\frac{2\sqrt{2}}{3}$

B $\frac{4}{3}$

C $\frac{3}{2}$

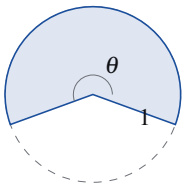
D $\sqrt{2}$

E 1

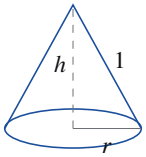
F 2

A sector of angle θ radians is cut from a disc of radius 1. The two straight edges of the sector are joined together, without overlap, to form the curved surface of a right circular cone.

Find the value of θ for which the volume of the cone is greatest.



sector cut from the disc



the cone

A $\frac{2\pi\sqrt{6}}{3}$

B $\frac{2\pi\sqrt{2}}{3}$

C $\frac{4\pi}{3}$

D $\pi\sqrt{2}$

E $\frac{2\pi\sqrt{3}}{3}$

F $\frac{3\pi}{2}$

The function f is defined for every real number except 0 and 1, and satisfies

$$f(x)+f\left(\frac{x-1}{x}\right)=1+x$$

for every such x. Find $f(2)$.

A $\frac{9}{4}$

B $-\frac{3}{4}$

C 1

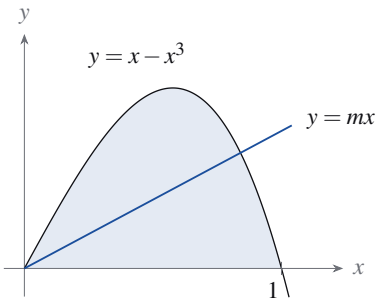
D $\frac{1}{2}$

E $\frac{3}{2}$

F $\frac{3}{4}$

The region R is enclosed between the curve $y = x - x^3$ and the x-axis, for $0 \leq x \leq 1$. The line $y = mx$, where $m > 0$, divides R into two regions of equal area.

Find m.



A $\dfrac{\sqrt{2}-1}{2}$

B $\dfrac{2-\sqrt{2}}{2}$

C $\dfrac{1}{2}$

D $1 - \dfrac{1}{\sqrt[3]{2}}$

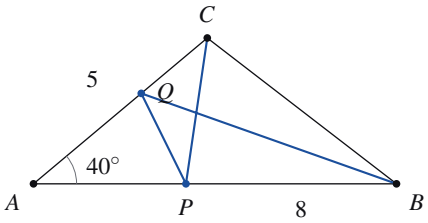
E $\dfrac{2-\sqrt[3]{2}}{2}$

F $\dfrac{3}{4}$

In triangle ABC the angle at A is $40°$, and $AB = 8$, $AC = 5$. The point P lies on the side AB and the point Q lies on the side AC.

Find the smallest possible value of

$$BQ + QP + PC.$$



A 7

B $\sqrt{89}$

C $\sqrt{109}$

D 12

E $\sqrt{129}$

F 13

Answer key

MARK BEFORE READING ON

Q	Ans	Value	Q	Ans	Value
1	**F**	$\frac{13\sqrt{10}}{10}$	11	**F**	$-2 < k < 6$
2	**A**	4	12	**D**	8640
3	**D**	-42	13	**E**	100
4	**B**	4	14	**E**	42
5	**E**	4	15	**B**	23
6	**C**	$\frac{42}{5}$	16	**D**	$\sqrt{2}$
7	**G**	$\frac{1}{2}$	17	**A**	$\frac{2\pi\sqrt{6}}{3}$
8	**D**	2	18	**F**	$\frac{3}{4}$
9	**C**	$\lvert S\rvert < 2\sqrt{2}-2$	19	**B**	$\frac{2-\sqrt{2}}{2}$
10	**B**	6	20	**E**	$\sqrt{129}$

Mark honestly, then read the solution to every question you did not get out cleanly, including the ones you guessed correctly.

Question 1 *answer F*

The identity that does the work. For positive x, y,

$$x^{\log_3 y} = 3^{(\log_3 x)(\log_3 y)} = y^{\log_3 x},$$

so both of the given expressions are powers of 3 with exponents built from $u = \log_3 x$ and $v = \log_3 y$. The first equation gives $3^{uv} = 27$, so $uv = 3$. For the second,

$$(xy)^{\log_3 x} = x^{\log_3 x} y^{\log_3 x} = 3^{u^2} \cdot 3^{uv} = 3^{u^2+uv},$$

so $u^2 + uv = 13$ and hence $u^2 = 10$. Since $x > 1$ we have $u > 0$, so $u = \sqrt{10}$ and $v = 3/\sqrt{10}$. Therefore

$$\log_3(xy) = u + v = \sqrt{10} + \frac{3}{\sqrt{10}} = \frac{10+3}{\sqrt{10}} = \frac{13\sqrt{10}}{10}.$$

Traps. $\sqrt{10}$ is $\log_3 x$ and $\frac{3\sqrt{10}}{10}$ is $\log_3 y$: both are answers to a question that was not asked. Reading the first equation as $3^{u+v} = 27$ gives 3. Note that the condition $x > 1$ is needed: $u = -\sqrt{10}$ also satisfies the equations, and gives $-\frac{13\sqrt{10}}{10}$.

Question 2 *answer A*

The angle in the corner. The chords OA and OB lie along the two axes, so $\angle AOB = 90^\circ$. An inscribed angle of 90° stands on a diameter, so AB is a diameter of the circle. Hence

$$a^2 + b^2 = (2r)^2 = 65^2 = 4225.$$

Counting the representations. We need $a^2 + b^2 = 4225$ with $0 < a < b$ integers. Since $65^2 = (5 \cdot 13)^2$ and both 5 and 13 are sums of two squares, several representations exist; checking $a = 1, \ldots, 45$ gives exactly

$$16^2 + 63^2, \qquad 25^2 + 60^2, \qquad 33^2 + 56^2, \qquad 39^2 + 52^2,$$

together with $0^2 + 65^2$, which is excluded because a must be positive. So there are 4 circles.

Traps. Counting ordered pairs (a, b) without the condition $a < b$ gives 8; including $0^2 + 65^2$ as well gives 5 or 10. Note that $a = b$ is impossible, since $4225/2$ is not a square.

Question 3 *answer D*

Evaluate at $x = 1$. If the roots are r_1, r_2, r_3 then

$$P(1) = 1 + a + b + c = 1 - 2026 = -2025,$$

and also $P(1) = (1 - r_1)(1 - r_2)(1 - r_3) = -(r_1 - 1)(r_2 - 1)(r_3 - 1)$. Hence

$$(r_1 - 1)(r_2 - 1)(r_3 - 1) = 2025 = 3^4 \cdot 5^2.$$

In particular every $r_i \geq 2$.

Maximising a. Since $a = -(r_1 + r_2 + r_3)$, the largest a comes from the smallest possible root sum. Writing $u_i = r_i - 1$, we need to minimise $u_1 + u_2 + u_3$ subject to $u_1 u_2 u_3 = 2025$. By AM–GM the sum is at least $3\sqrt[3]{2025} > 37.9$, so at least 38; and running through the divisors of 2025 shows the minimum is 39, attained only at $\{u_i\} = \{9, 15, 15\}$. Hence $\{r_i\} = \{10, 16, 16\}$, the root sum is 42, and

$$a_{\max} = -42.$$

The search has to be complete. Ordered by root sum, the best few factorisations are

$$\{9, 15, 15\} \to 42, \qquad \{9, 9, 25\} \to 46, \qquad \{5, 15, 27\} \to 50, \qquad \{3, 25, 27\} \to 58,$$

so a search that stops at the first "balanced-looking" triple it finds returns -46 or -50 instead. The AM–GM bound is what tells you when to stop: once a sum of 39 is found and 38 is known to be impossible, the search is over.

Traps. Forgetting the shift $r_i = u_i + 1$ and answering $-(u_1 + u_2 + u_3) = -39$ is the commonest slip. Minimising a instead of maximising it means taking the largest root sum, $2025 + 1 + 1$ shifted up by 3, which gives the offered -2030. Answering with the sum of the roots rather than with a gives 42.

Question 4 *answer B*

Difference the defining relation. Let $S_n = u_1 + \cdots + u_n = \frac{n(n+2)}{3}u_n$. For $n \geq 2$,

$$u_n = S_n - S_{n-1} = \frac{n(n+2)}{3}u_n - \frac{(n-1)(n+1)}{3}u_{n-1},$$

so $3u_n = n(n+2)u_n - (n^2-1)u_{n-1}$, that is

$$(n^2+2n-3)u_n = (n^2-1)u_{n-1} \implies (n+3)(n-1)u_n = (n+1)(n-1)u_{n-1}.$$

For $n \geq 2$ we may cancel $n-1$, leaving the clean recurrence

$$u_n = \frac{n+1}{n+3}\,u_{n-1}.$$

Telescoping. With $u_1 = 1$,

$$u_n = \prod_{k=2}^{n} \frac{k+1}{k+3} = \frac{3\cdot 4\cdots(n+1)}{5\cdot 6\cdots(n+3)} = \frac{4\cdot 3}{(n+2)(n+3)} = \frac{12}{(n+2)(n+3)},$$

which indeed gives $u_1 = 12/12 = 1$. Finally

$$\sum_{n=1}^{\infty} u_n = 12\sum_{n=1}^{\infty}\left(\frac{1}{n+2} - \frac{1}{n+3}\right) = \frac{12}{3} = 4.$$

Remark. The cancellation of $n-1$ is legitimate only for $n \geq 2$; at $n = 1$ the relation is the given $u_1 = u_1$ and carries no information, which is why the initial value must be supplied separately. Stopping the telescoping one term early gives $\frac{12}{4} = 3$.

Question 5 *answer E*

Reduce to one variable. Since $\sin^2 x + \cos^2 x = 1$,

$$\sin^6 x + \cos^6 x = 1 - 3\sin^2 x\cos^2 x = 1 - \tfrac{3}{4}\sin^2 2x.$$

Writing $s = \sin 2x$, the equation becomes $1 - \frac{3}{4}s^2 = \frac{1}{4}(s+1)$, i.e.

$$4 - 3s^2 = s + 1 \implies 3s^2 + s - 3 = 0 \implies s = \frac{-1 \pm \sqrt{37}}{6}.$$

The range check is the question. Since $\sqrt{37} \approx 6.08$, the two roots are $s \approx 0.847$ and $s \approx -1.180$. Only the first lies in $[-1, 1]$, so exactly one value of s survives.

Counting x. As x runs over $[0, 2\pi]$, the angle $2x$ runs over $[0, 4\pi]$ — two full turns. The equation $\sin 2x = s$ with $0 < s < 1$ therefore has $2 \times 2 = 4$ solutions.

Traps. Forgetting that $2x$ sweeps twice round gives 2; keeping the rejected root as well gives 8; solving for $\sin^2 2x$ and taking both signs gives 6.

Question 6 *answer C*

The symmetry trick. Let I be the integral and substitute $x \mapsto -x$, which maps $[-2,2]$ to itself:

$$I = \int_{-2}^{2} \frac{x^4+1}{1+3^{-x}}\,dx = \int_{-2}^{2} \frac{3^x\left(x^4+1\right)}{3^x+1}\,dx,$$

using that x^4+1 is even. Adding this to the original expression for I, the denominators combine:

$$2I = \int_{-2}^{2} \frac{\left(x^4+1\right)\left(1+3^x\right)}{1+3^x}\,dx = \int_{-2}^{2} \left(x^4+1\right)dx.$$

Finish. $\displaystyle\int_{-2}^{2}\left(x^4+1\right)dx = 2\left(\frac{32}{5}+2\right) = \frac{84}{5}$, so $I = \dfrac{42}{5}$.

Traps. $\frac{84}{5}$ is the value of $2I$ — the commonest slip. $\frac{64}{5}$ and $\frac{32}{5}$ come from dropping the $+1$, and $\frac{21}{5}$ from halving twice. No antiderivative of the integrand is needed, or available.

Question 7 *answer G*

Set up the two states. Each roll is a 6 (probability $\frac{1}{6}$), odd (probability $\frac{1}{2}$), or an even number other than 6, i.e. 2 or 4 (probability $\frac{1}{3}$). Let p be the required probability starting fresh, and q the required probability immediately after an odd roll. From a fresh position,

$$p = \tfrac{1}{6} + \tfrac{1}{2}q + \tfrac{1}{3}p,$$

and from just after an odd roll a further odd number ends the game unsuccessfully, so

$$q = \tfrac{1}{6} + \tfrac{1}{3}p.$$

Solve. Substituting the second into the first,

$$p = \tfrac{1}{6} + \tfrac{1}{2}\left(\tfrac{1}{6} + \tfrac{1}{3}p\right) + \tfrac{1}{3}p = \tfrac{1}{4} + \tfrac{1}{2}p \implies p = \tfrac{1}{2}.$$

Remark. The game ends with probability 1, so no separate convergence argument is needed: two odd rolls in a row occur eventually unless a 6 arrives first. The value $\frac{1}{2}$ is exact, not an approximation — a pleasant surprise, and precisely why the neighbouring options $\frac{4}{9}$ and $\frac{5}{9}$ are offered. Forgetting that 6 is itself even (and so cannot also count as "even other than 6") gives $\frac{5}{9}$; treating the odd-run condition as needing two odds in total rather than consecutively gives $\frac{3}{7}$.

Question 8 *answer D*

Find the intersections. The circle through the origin with centre $(0,c)$ is $x^2+(y-c)^2=c^2$, i.e. $x^2+y^2-2cy=0$. Substituting $y=x^2$, so that $x^2=y$, gives

$$y+y^2-2cy=0 \implies y\big(y-(2c-1)\big)=0.$$

So the intersections are the origin ($y=0$) and, when $2c-1>0$, the two points with $y=2c-1$, namely $\big(\pm\sqrt{2c-1},\,2c-1\big)$. There are exactly three intersection points precisely when $2c-1>0$.

Impose the equilateral condition. Put $t=2c-1>0$. The horizontal side has length $2\sqrt{t}$, and the distance from the origin to $\big(\sqrt{t},t\big)$ is $\sqrt{t+t^2}$. Equating the squares,

$$4t=t+t^2 \implies t=3 \implies c=2.$$

Check. The points are $(0,0)$ and $\big(\pm\sqrt{3},3\big)$; every side has length $2\sqrt{3}$, and each point satisfies $x^2+(y-2)^2=4$.

Trap. $c=\frac{1}{2}$ is the threshold at which the two extra points merge into the origin — the answer to "when is the origin the only intersection", not to the question asked.

Question 9 *answer C*

Read the graph. Splitting at 0,

$$f(x)=\begin{cases} x^2-4x, & x\geq 0,\\ -x^2-4x, & x<0.\end{cases}$$

On $x\geq 0$ this is a parabola falling from $f(0)=0$ to a minimum $f(2)=-4$ and then rising. On $x<0$ it is $4-(x+2)^2$, rising from $-\infty$ to a maximum $f(-2)=4$ and falling back to 0 at $x=0$. So f has local maximum value 4 and local minimum value -4, and there are exactly three distinct roots precisely when $-4<k<4$.

Compute the sum in each case. For $0<k<4$ the two roots with $x<0$ satisfy $x^2+4x+k=0$, so they sum to -4; the remaining root is the positive solution of $x^2-4x=k$, namely $2+\sqrt{4+k}$. Hence

$$S=-4+2+\sqrt{4+k}=\sqrt{4+k}-2.$$

For $-4<k<0$ the mirror-image argument gives $S=2-\sqrt{4-k}$. Both formulas give $S=0$ at $k=0$.

The range. As k runs over $(0,4)$, $\sqrt{4+k}-2$ runs over $\big(0,\,2\sqrt{2}-2\big)$; as k runs over $(-4,0)$, $2-\sqrt{4-k}$ runs over $\big(2-2\sqrt{2},\,0\big)$. Together with $S=0$,

$$2-2\sqrt{2}<S<2\sqrt{2}-2,$$

an interval of half-width $2\sqrt{2}-2\approx 0.828$. Both endpoints are excluded, since $k=\pm 4$ produces a repeated root and only two distinct solutions.

Traps. $-4<S<4$ is the range of k, not of S; $0\leq S<2\sqrt{2}-2$ keeps only the case $k>0$.

Question 10 *answer B*

Reduce the divisibility. Because $n^3 - 1 = (n-1)\left(n^2+n+1\right)$, we have $n^3 \equiv 1 \pmod{n^2+n+1}$, and therefore

$$n^3 + 272 \equiv 273 \pmod{n^2+n+1}.$$

So the condition is exactly $n^2+n+1 \mid 273$. This is what makes the problem finite: the divisor is bounded, so only finitely many n can work.

Run through the divisors. $273 = 3 \cdot 7 \cdot 13$, with divisors $1, 3, 7, 13, 21, 39, 91, 273$. Solving $n^2+n+1 = d$ for each gives

$$d = 3 \to n = 1, \qquad d = 7 \to n = 2, \qquad d = 13 \to n = 3,$$

$$d = 21 \to n = 4, \qquad d = 91 \to n = 9, \qquad d = 273 \to n = 16.$$

For $d = 39$ the equation $n^2+n-38 = 0$ has discriminant 153, which is not a perfect square, so there is no integer solution; and $d = 1$ gives $n = 0$, which is not positive. So $n \in \{1, 2, 3, 4, 9, 16\}$ and the answer is 6.

Trap. Searching by hand and stopping at $n = 10$ finds only $1, 2, 3, 4, 9$ and gives 5; without the divisor bound there is no reason to believe the list has ended, which is the whole point of the question.

Question 11 *answer F*

The shape of the cubic. $f'(x) = 3x^2 - 6x$, so f has a local maximum $f(0) = k$ and a local minimum $f(2) = k-4$. For a real constant λ, the equation $f(x) = \lambda$ has three distinct roots if $k-4 < \lambda < k$, exactly two distinct roots if $\lambda = k$ or $\lambda = k-4$, and one root otherwise.

Combine the two levels. $|f(x)| = 2$ means $f(x) = 2$ or $f(x) = -2$, and the two root sets are disjoint because f cannot take both values at one point. Writing $N(\lambda)$ for the number of distinct roots at level λ, we need $N(2) + N(-2) = 4$. Since $N \in \{1, 2, 3\}$, the possibilities are $3+1$, $1+3$ and $2+2$:

- $N(2) = 3$ needs $k-4 < 2 < k$, i.e. $2 < k < 6$; then $-2 < k-4$, so $N(-2) = 1$. Valid.
- $N(-2) = 3$ needs $k-4 < -2 < k$, i.e. $-2 < k < 2$; then $2 > k$, so $N(2) = 1$. Valid.
- $N(2) = N(-2) = 2$ needs $k = 2$ and $k-4 = -2$ simultaneously, i.e. $k = 2$. Valid.

The three cases fit together into the single interval $-2 < k < 6$.

The endpoints. At $k = 6$ the level 2 passes through the local minimum, giving $N(2) = 2$ and $N(-2) = 1$: only three solutions. Likewise $k = -2$ gives three. So both endpoints fail.

Trap. The value $k = 2$ looks exceptional — it is the one case where neither level gives three roots — but it still yields $2+2 = 4$ solutions, so excluding it (option E) is wrong.

Question 12 *answer D*

Split into chains. The only constraint links m and $3m$, so group $\{1,\dots,15\}$ into maximal chains under multiplication by 3:

$$\{1,3,9\},\quad \{2,6\},\quad \{4,12\},\quad \{5,15\},\quad \{7\},\{8\},\{10\},\{11\},\{13\},\{14\}.$$

No constraint runs between different chains, so the total is the product of the counts for each chain.

Count each chain. For a chain we must choose a subset containing no two adjacent elements. A chain of length 1 gives 2 subsets, a chain of length 2 gives 3, and a chain of length 3 gives 5 (all subsets of $\{1,3,9\}$ except those containing $\{1,3\}$ or $\{3,9\}$: that is $\emptyset,\{1\},\{3\},\{9\},\{1,9\}$). These are consecutive Fibonacci numbers, as they must be.

Multiply. $5\times 3^3\times 2^6 = 5\times 27\times 64 = 8640$.

Traps. Counting $\{1,3,9\}$ as 4 (forgetting that $\{1,9\}$ is allowed, since $9\neq 3\cdot 1$) gives 6912; treating the three two-element chains as contributing 2 each gives 2560.

Question 13 *answer E*

Rationalise. Factor the denominator:

$$n\sqrt{n+1}+(n+1)\sqrt{n} = \sqrt{n}\sqrt{n+1}\left(\sqrt{n}+\sqrt{n+1}\right),$$

so, multiplying above and below by $\sqrt{n+1}-\sqrt{n}$,

$$\frac{1}{n\sqrt{n+1}+(n+1)\sqrt{n}} = \frac{\sqrt{n+1}-\sqrt{n}}{\sqrt{n}\sqrt{n+1}} = \frac{1}{\sqrt{n}}-\frac{1}{\sqrt{n+1}}.$$

Telescope. Hence

$$\sum_{n=1}^{N}\frac{1}{n\sqrt{n+1}+(n+1)\sqrt{n}} = 1-\frac{1}{\sqrt{N+1}}.$$

Solve the inequality — carefully.

$$1-\frac{1}{\sqrt{N+1}}>\frac{9}{10} \iff \frac{1}{\sqrt{N+1}}<\frac{1}{10} \iff \sqrt{N+1}>10 \iff N>99.$$

The inequality is strict, and at $N=99$ the sum equals $1-\frac{1}{10}=\frac{9}{10}$ exactly. So the smallest integer that works is $N=100$.

Trap. 99 is the answer to the same question with $\geq$ in place of $>$, and is the option most students choose.

Question 14 *answer E*

Halve the chords. Let O_1, O_2 be the centres and let M_1, M_2 be the feet of the perpendiculars from O_1, O_2 to the line PAQ. A perpendicular from the centre bisects a chord, so M_1 is the midpoint of AP and M_2 is the midpoint of AQ. Since A lies between P and Q, the points M_1 and M_2 lie on opposite sides of A, and

$$M_1M_2 = \tfrac{1}{2}AP + \tfrac{1}{2}AQ = \tfrac{1}{2}PQ.$$

Bound it. M_1M_2 is the length of the projection of O_1O_2 onto the line, so $M_1M_2 \le O_1O_2 = 21$, with equality exactly when the line is parallel to O_1O_2. Hence

$$PQ = 2M_1M_2 \le 2\,O_1O_2 = 42.$$

Why it is attained. The line through A parallel to O_1O_2 does meet each circle a second time and does have A between the two second points, so 42 is achieved and is not merely an upper bound. Concretely, place O_1 at the origin and O_2 at $(21,0)$. Then $A = (6,8)$, since $6^2+8^2 = 100$ and $(6-21)^2+8^2 = 225+64 = 289$. The horizontal line $y = 8$ meets the first circle where $x^2 = 36$, giving $P = (-6,8)$, and meets the second where $(x-21)^2 = 225$, giving $Q = (36,8)$. So

$$PQ = 36 - (-6) = 42.$$

Traps. 16 is the length of the common chord (the foot of the perpendicular from O_1 is at distance 6 along the line of centres, and the half-chord is $\sqrt{10^2-6^2} = 8$). The value 27 is the sum of the radii and 21 the distance between the centres — both are natural guesses, and both are wrong: the bound involves twice the distance between the centres, not the radii at all. The option $2\sqrt{445} \approx 42.19$ sits just above the true maximum, so estimating a diagram cannot separate it from 42.

Question 15 *answer B*

Set up. Let the list, sorted, be $x_1 \le x_2 \le \cdots \le x_7$ of positive integers, with $x_1 + \cdots + x_7 = 49$ and $x_4 = 7$. We want $x_7 - x_1$ as large as possible, so we want x_1 small and x_7 large, which means making $x_2, \ldots, x_6$ as small as their constraints allow.

The mode is the binding constraint. Since $x_4 = 7$ and 7 is the unique mode, 7 must occur strictly more often than any other value. Suppose 7 occurs exactly three times: then it occupies x_4, x_5, x_6 (it cannot occupy a position below x_4 without forcing $x_4 > 7$), and every other value may occur at most twice. Taking $x_1 = x_2 = 1$ and $x_3 = 2$ — we cannot take $x_3 = 1$, since three 1s would tie the mode — gives

$$1+1+2+7+7+7+x_7 = 49 \implies x_7 = 24,$$

and the range is $24 - 1 = 23$.

Nothing does better. If 7 occurs only twice, all other values are distinct, so $x_1, x_2, x_3 \ge 1, 2, 3$ and $x_6 \ge 8$, giving $x_7 \le 49 - (1+2+3+7+7+8) = 21$ and a range of at most 20. If 7 occurs four or more times it fills $x_4, \ldots, x_7$, so $x_7 = 7$ and the range is at most 6. Hence 23 is the maximum.

Traps. 24 is the largest element, not the range; 20 is the best available when 7 occurs only twice; 41 ignores the mode condition entirely.

Question 16 *answer D*

Guess the shape of the answer first. The denominator $ab+bc=b(a+c)$ is symmetric in a and c, so at the minimum we expect $a=c$. Setting $a=c$ and scaling $b=1$, the expression becomes

$$g(a)=\frac{2a^2+1}{2a},$$

and $g'(a)=0$ gives $a=\frac{1}{\sqrt{2}}$, with $g\left(\frac{1}{\sqrt{2}}\right)=\sqrt{2}$.

Prove the bound. Split b^2 into two halves and apply AM–GM twice:

$$a^2+\tfrac{1}{2}b^2\geq 2\sqrt{\tfrac{1}{2}}\,ab=\sqrt{2}\,ab, \qquad c^2+\tfrac{1}{2}b^2\geq\sqrt{2}\,bc.$$

Adding, $a^2+b^2+c^2\geq\sqrt{2}(ab+bc)$ for all positive a,b,c, so the ratio is always at least $\sqrt{2}$; and equality holds when $a=c=\frac{b}{\sqrt{2}}$, which is attainable. The minimum is $\sqrt{2}$.

Why the obvious test fails. Substituting $a=b=c$ — the first thing most candidates try — gives $\frac{3}{2}$, which is offered as option C. Since $\frac{3}{2}>\sqrt{2}$, that test produces a value that is not the minimum and is not even close enough to identify it. The splitting $b^2=\frac{1}{2}b^2+\frac{1}{2}b^2$ is forced: any other split makes the two AM–GM constants differ, and the weaker of them governs the bound.

Question 17 *answer A*

Build the cone. The arc of the sector becomes the circumference of the base, so if the base radius is r then $2\pi r=\theta\cdot 1$, giving $r=\frac{\theta}{2\pi}$. The slant height is the radius of the disc, namely 1, so the height is $h=\sqrt{1-r^2}$ and

$$V=\tfrac{1}{3}\pi r^2 h=\tfrac{1}{3}\pi r^2\sqrt{1-r^2}.$$

Maximise. It is easiest to maximise $V^2\propto r^4\left(1-r^2\right)$. Differentiating,

$$\frac{d}{dr}\left(r^4-r^6\right)=4r^3-6r^5=2r^3\left(2-3r^2\right),$$

which vanishes for $r^2=\frac{2}{3}$, a maximum. Hence $r=\sqrt{\frac{2}{3}}=\frac{\sqrt{6}}{3}$ and

$$\theta=2\pi r=\frac{2\pi\sqrt{6}}{3}.$$

Check it is a legal sector. $\frac{2\pi\sqrt{6}}{3}\approx 5.13<2\pi\approx 6.28$, so the sector is a genuine proper sector of the disc.

Traps. Reading $r^2=\frac{2}{3}$ as $r=\frac{2}{3}$ gives $\frac{4\pi}{3}$; so does using $h=1-r$ in place of $h=\sqrt{1-r^2}$. The option $\frac{2\pi\sqrt{2}}{3}$ is what appears when $\sqrt{\frac{2}{3}}$ is "simplified" to $\frac{\sqrt{2}}{3}$. Maximising $r^2\sqrt{1-r^2}$ as though the exponent of the bracket were 1 gives $r^2=\frac{1}{2}$ and $\pi\sqrt{2}$. The remaining two options sit either side of the answer and cannot be separated from it by estimating the picture.

Question 18 *answer F*

The substitution has order three. Let $g(x)=\dfrac{x-1}{x}$. Then

$$g\big(g(x)\big)=\frac{\frac{x-1}{x}-1}{\frac{x-1}{x}}=\frac{-1/x}{(x-1)/x}=\frac{1}{1-x},\qquad g\big(g(g(x))\big)=x.$$

So applying g three times returns to the start, and the three equations obtained from x, $g(x)$ and $g(g(x))$ close up.

Use the cycle through $x=2$. Here $2\to\frac{1}{2}\to-1\to 2$, so writing $A=f(2)$, $B=f\big(\frac{1}{2}\big)$, $C=f(-1)$ the given relation yields

$$A+B=1+2=3,\qquad B+C=1+\tfrac{1}{2}=\tfrac{3}{2},\qquad C+A=1+(-1)=0.$$

Solve. Adding all three gives $2(A+B+C)=\frac{9}{2}$, so $A+B+C=\frac{9}{4}$. Subtracting the middle equation,

$$f(2)=A=\tfrac{9}{4}-\tfrac{3}{2}=\tfrac{3}{4}.$$

Traps. $\frac{9}{4}$ is $A+B+C$, and also the value of $f\big(\frac{1}{2}\big)$; $-\frac{3}{4}$ is $f(-1)$. Solving only two of the three equations leaves one unknown free and tempts the answer $\frac{3}{2}$.

Question 19 *answer B*

Total area. $\displaystyle\int_0^1\big(x-x^3\big)dx=\tfrac{1}{2}-\tfrac{1}{4}=\tfrac{1}{4}$, so each part must have area $\frac{1}{8}$.

Where the line cuts the curve. Solving $x-x^3=mx$ gives $x=0$ or $x^2=1-m$. For a genuine crossing inside the region we need $0<m<1$; put $x_0=\sqrt{1-m}$.

The upper part. The region between the curve and the line runs from 0 to x_0:

$$\int_0^{x_0}\big((1-m)x-x^3\big)dx=\frac{(1-m)x_0^2}{2}-\frac{x_0^4}{4}=\frac{(1-m)^2}{2}-\frac{(1-m)^2}{4}=\frac{(1-m)^2}{4}.$$

Set it equal to half.

$$\frac{(1-m)^2}{4}=\frac{1}{8}\implies(1-m)^2=\frac{1}{2}\implies 1-m=\frac{1}{\sqrt{2}},$$

taking the positive root because $m<1$. Hence

$$m=1-\frac{\sqrt{2}}{2}=\frac{2-\sqrt{2}}{2}\approx 0.293.$$

Remark. The algebraic solution $m=1+\frac{\sqrt{2}}{2}$ is rejected: it gives $1-m<0$, so the line never meets the curve again in $x>0$ and cuts off nothing. The cube-root options are what appears if the area is mistakenly taken to scale like $(1-m)^3$.

Question 20 *answer E*

Unfold the path. Reflect C in the line AB to get C', and reflect B in the line AC to get B'. Reflection preserves distance to points of the mirror line, so for any P on AB and Q on AC,

$$PC = PC', \qquad QB = QB',$$

and therefore

$$BQ + QP + PC = B'Q + QP + PC' \;\geq\; B'C',$$

since the right-hand side is a straight-line distance and the left-hand side is a path from B' to C'.

Compute $B'C'$. Take ray AB at angle 0° and ray AC at angle 40°. Then C' sits at angle -40° with $AC' = 5$, and B' at angle 80° with $AB' = 8$. So

$$\angle B'AC' = 40^\circ + 40^\circ + 40^\circ = 120^\circ,$$

and the cosine rule gives

$$B'C'^2 = 8^2 + 5^2 - 2 \cdot 8 \cdot 5 \cos 120^\circ = 64 + 25 + 40 = 129.$$

It is attained. Because $\angle B'AC' = 120^\circ < 180^\circ$, the segment $B'C'$ genuinely crosses ray AB and then ray AC, in that order, giving admissible points P and Q on the two sides; one finds $AP \approx 3.12 < 8$ and $AQ \approx 3.44 < 5$, so both lie on the sides and not on their extensions. Hence the minimum is $\sqrt{129}$.

Traps. Reflecting only once leaves $\angle B'AC' = 80^\circ$ and a value that is not offered; taking $\angle B'AC' = 90^\circ$ gives $\sqrt{89}$. The value 13 is what the same construction returns when $\angle A = 60^\circ$, where $\angle B'AC' = 180^\circ$ and the three reflected copies fall into a straight line — a genuinely different problem, and the reason the angle here is 40°.

Provenance and syllabus notes

This set is a prediction, not an anthology. Every question was written after reading the 2024, 2025 and 2026 papers, and the shape of the paper is copied from them rather than chosen. Four features of those papers drive almost all of it. The first is that a Paper 1 question has a single answer and no reasoning apparatus: there is not one necessary-and-sufficient item here, and not one answer lattice, because the recent Paper 1s carry essentially none. The second is the option count, which sits between four and eight and is most often six; this paper uses six for fourteen questions and seven for three. The third is the terseness of the stems: the real paper gives the configuration and asks for a number, with no scaffolding, no "hence" and no part (a). The fourth is that the difficulty is not ordered. Questions 5, 6 and 18 are easier than the paper's average and sit in the middle of it, exactly as an easy question sits at 18 on a real paper.

On syllabus

Every question sits inside the TMUA Paper 1 specification. The following were checked explicitly.

- No differentiation of $\ln x$ or e^x, and no integration of either, appears anywhere. Question 6 contains 3^x, but it is never differentiated or integrated: the whole question is the substitution $x \mapsto -x$, and the exponential cancels.
- No modular arithmetic is assumed. Question 10 is a divisibility argument built from the factorisation of $n^3 - 1$, and Question 3 from evaluating a polynomial at $x = 1$.
- Series are limited to arithmetic, geometric and telescoping sums. Questions 4 and 13 telescope; neither needs a convergence test beyond the fact that the tail of a telescoping sum goes to zero.
- Probability is limited to independence and conditional probability. Question 7 is a two-state recurrence solved by simultaneous equations, which is on the specification; no Markov theory is needed or used.
- Question 16 uses AM–GM in the two-variable form $u^2 + v^2 \geq 2uv$, which is the arithmetic-geometric inequality the specification lists.

On the answer options

Every option list was built after the solution, from the wrong answers the question actually produces. Question 1 offers $\log_3 x$ and $\log_3 y$ beside $\log_3(xy)$; Question 3 offers -46 and -50, which are the next two factorisations of 2025 in order of root sum, and -39, which is the answer of a candidate who forgets the shift; Question 6 offers $2I$; Question 13 offers the answer to the same question with $\geq$ in place of $>$; Question 14 offers the sum of the radii, the distance between the centres, the common chord, and $2\sqrt{445}$, which is close enough to 42 that no drawing can separate them; Question 15 offers the largest element rather than the range. In no question does substituting an option into the stem decide anything, and in no question is the answer the value a candidate reaches by testing the symmetric case: Question 16 in particular has $\frac{3}{2}$, the value at $a = b = c$, as a distractor.

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